

Microeconomics for Finance

Equilibrium in Markets for Securities

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The Arrow-Debreu Economy

Arrow-Debreu Theory

The Arrow-Debreu theory has been developed during the 1950/60's by Kenneth Arrow (Nobel Prize 1972) and Gerard Debreu (Nobel Prize 1983)

Bibliography.

Gerard Debreu, *Theory of Value : An Axiomatic Analysis of Economic Equilibrium*, New Haven : Yale University Press, 1959.

Kenneth Arrow, "The Role of Securities in the Optimal Allocation of Risk-Bearing", *Review of Economic Studies* Vol.31 (April 1964) : pp.91-96.

Arrow-Debreu Theory

Arrow-Debreu offer an equilibrium theory of securities/asset prices.

Arrow-Debreu theory is key to understand the modern techniques of asset pricing, in particular in financial mathematics.

Arrow-Debreu Theory

What is the price of a risky payment $\tilde{D}$?

First, divide it in several components, $D_1, D_2, \dots, D_N$, paid in each of the N different states of the world that can occur each with probability π_n .

$$E[\tilde{D}] = \pi_1 D_1 + \pi_2 D_2 + \dots + \pi_N D_N$$

Introduce N securities, where security n pays 1\$ if the state n is realized, and 0 otherwise. Denote q_n the security's price. Then, the price of the risky payment should be :

$$P = q_1 D_1 + q_2 D_2 + \dots + q_N D_N$$

In this class we focus on the $N = 2$ states case.

Assumptions

- ① There are two dates $t = 0$ and $t = 1$. At $t = 1$, there are 2 states of the world, each can be realized with probability $\pi_1 = \pi$ and $\pi_2 = 1 - \pi$.
- ② There are 2 agents, A and B , who can consume at $t = 0$ and $t = 1$, and obtains an expected utility

$$u_i(c_0^i) + \beta (\pi_1 u_i(c_1^i) + \pi_2 u_i(c_2^i)) ,$$

where c_0^i , c_1^i and c_2^i are the consumption at $t = 0$ and $t = 1$ in each state, β is the time discount factor, and u_i is increasing and strictly concave.

- ③ Agent $i = A, B$ is “endowed” with consumption units at $t = 0$, and in each state of the world at $t = 1$: $w_0^i, w_n^i, n \in \{1, 2\}$.
- ④ A consumption unit is perishable and cannot be transferred from $t = 0$ to $t = 1$, or across states.

Financial Markets

- Each agents faces the risk of consumption fluctuation over times and across states of the world.
- Agents would gain by trading their endowments with each other to smooth their consumption and share their risk.
- Via financial markets, agents can “transfer” consumption units across times and states.

Financial Markets :

- There are 3 **Arrow-Debreu** securities that can be traded in competitive markets at $t = 0$.
- One unit of security 0 gives one unit of consumption at $t = 0$. Its price is $p_0 = 1$. This is the market for current consumption units.
- One unit of security $n \in \{1, 2\}$ gives one unit of consumption at $t = 1$, in state n . Its price is q_n .
- The net-supply of each of these securities is zero.

Competitive Equilibrium

Equilibrium Conditions

- Let z_0^i and z_n^i , $n \in \{1, 2\}$, agent i demands for A-D securities.
- His consumption at each period and each state is :

$$c_0^i = w_0^i + z_0^i, \quad c_n^i = w_n^i + z_n^i, \quad n \in \{1, 2\}$$

- Financial markets must clear :

$$z_0^A + z_0^B = 0, \quad \text{and } \forall n \quad z_n^A + z_n^B = 0$$

→ Equivalent to market clearing conditions for consumption units :

$$c_0^A + c_0^B = w_0^A + w_0^B, \quad \text{and } \forall n \quad c_n^A + c_n^B = w_n^A + w_n^B$$

- Each agent has a budget constraint :

$$w_0^i + q_1 w_1^i + q_2 w_2^i = c_0^i + q_1 c_1^i + q_2 c_2^i \iff z_0^i + q_1 z_1^i + q_2 z_2^i = 0$$

Walras Law

- By summing all the agents' budget constraints, we obtain

$$(z_0^A + z_0^B) + q_1(z_1^A + z_1^B) + q_2(z_2^A + z_2^B) = 0$$

- If markets for securities $n \in \{1, 2\}$ clear, then the market for security 0 clears as well :

$$\forall n \ z_n^A + z_n^B = 0 \Rightarrow z_0^A + z_0^B = 0$$

Agents Optimization

- Agent i chooses a consumption plan that maximizes his utility under his budget constraint.
- The Lagrangian of this maximization program is

$$\mathcal{L}_i = u_i(c_0^i) + \beta (\pi_1 u_i(c_1^i) + \pi_2 u_i(c_2^i)) \\ + \lambda_i (w_0^i + q_1 w_1^i + q_2 w_2^i - c_0^i - q_1 c_1^i - q_2 c_2^i)$$

- First Order Conditions :

$$\frac{\partial \mathcal{L}_i}{\partial c_0^i} = u'_i(c_0^i) - \lambda_i = 0$$

$$\frac{\partial \mathcal{L}_i}{\partial c_n^i} = \beta \pi_n u'_i(c_n^i) - \lambda_i q_n = 0$$

Equilibrium

$$\forall i, q_n = \frac{\beta \pi_n u'_i(c_n^i)}{u'_i(c_0^i)}$$

The price of one unit of consumption in state n :

- is low when the probability π_n is low
- is low when the time discount factor is small
- is large when the level of consumption is low (because $u'' < 0$).

Social Optimum

In the Arrow-Debreu economy, the equilibrium is competitive. We can apply the Welfare Theorems.

First Welfare Theorem. *A competitive equilibrium is Pareto optimal.*

Perfect Risk Sharing

- Assume there is no aggregate risk at $t = 1$:

$$w_1 = w_2$$

- One can show that $c_1^i = c_2^i$ is obtained in the competitive equilibrium.
- The idiosyncratic risk (initial endowments) is fully diversified.
- The market allows for perfect risk sharing.

Asset Pricing

Law of One Price

- In addition of A-D securities, one can add a security (in zero-net supply) that pays D_n in state of the world n , at $t = 1$
- The price of this security can be determined by the *law of one price* :

$$p = q_1 D_1 + q_2 D_2$$

- If p was different, one could replicate the pay-off of the security with A-D securities, and exploit the price difference by taking long and short positions in the security and the replication portfolio.

Law of One Price

Proof. Denote z^i agent i demand for the new security. He maximizes

$$u_i(c_0^i) + \beta (\pi_1 u_i(c_1^i) + \pi_2 u_i(c_2^i))$$

$$\text{with } c_0^i = w_0^i + z_0^i, \quad c_n^i = w_n^i + z_n^i + z^i D_n, \quad n \in \{1, 2\}$$

under the budget constraint

$$z_0^i + q_1 z_1^i + q_2 z_2^i + p z^i = 0$$

The FOCs with respect to z_0^i , z_n^i 's and z^i are

$$u_i'(c_0^i) - \lambda_i = 0, \quad \beta \pi_n u_i'(c_n^i) - \lambda_i q_n = 0$$

$$\beta (\pi_1 u_i(c_1^i) D_1 + \pi_2 u_i(c_2^i) D_2) - \lambda_i p = 0$$

which implies

$$p = q_1 D_1 + q_2 D_2,$$

with q_n 's that would be the same without this redundant security.

Complete Markets

Definition. Markets are complete if any profile of risky payments can be replicated with existing financial securities.

In our case a risky payment v is :

- v_1 and v_2 at $t = 1$ in state 1 and 2.
- Obviously we can replicate this payment profile with the portfolio of Arrow-Debreu securities 1 and 2, (v_1, v_2) .

⇒ In the Arrow-Debreu economy, markets are complete.

Complete Markets

Assume that, instead of Arrow-Debreu securities 1 and 2, one can trade 2 securities a^1 and a^2 , with payment matrix, at $t = 1$:

$$A = \begin{pmatrix} a_1^1 & a_2^1 \\ a_1^2 & a_2^2 \end{pmatrix}$$

The payment of portfolio $z' = (z_1, z_2)'$ is $A'z$. Therefore,

A is full-ranked $\Leftrightarrow$ complete markets.

Asset pricing implications. When markets are complete, one can replicate the pay-offs of *synthetic* A-D securities, obtain their prices q_1 and q_2 , and price any additional security.